

Resources, Comparative Advantage, and Income Distribution

ECON 308 – International Economic Relations Chapter 4 (Specific Factors)

Muhebullah Karimzada

School of Economics
University of Northern British Columbia

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Opening Puzzle: Why Do Steelworkers Oppose Free Trade?

- In 2018 the US imposed a 25% tariff on Canadian steel. Hamilton steelworkers cheered. Canadian auto workers groaned.
- If the Ricardian model is right that both countries gain from trade, why does trade generate such fierce political opposition?
- The answer: the Ricardian model is *too simple*. When capital and land are fixed in specific sectors in the short run, trade creates **real winners and losers**.
- This chapter adds capital and land to the model — and suddenly we have a micro-foundation for understanding trade politics.

The Distributional Question

Trade raises **national income** but the gains are not shared equally. The Specific Factors Model explains which groups gain and which lose — and why they lobby so differently.

Learning Objectives

1. State the assumptions of the specific factors model and explain what “specific” means.
2. Derive the marginal product of labour schedule and identify the labour market equilibrium.
3. Predict the effect of a change in goods prices on wages, capital returns, and land returns.
4. Explain the magnification effect and why factor returns change more than prices.
5. Identify winners and losers from trade and apply the compensation principle.
6. Analyse the political economy of trade policy through the lens of factor specificity.
7. Discuss the effects of international labour mobility.

Model Setup: Three Factors, Two Sectors

Two goods:

- **Manufactures (M)**: capital-intensive.
- **Food (F)**: land-intensive.

Three factors:

- **Labour** L : **mobile** — can work in either sector.
- **Capital** K : **specific** to manufactures only.
- **Land** T : **specific** to food only.

Why “specific”?

In the short run, a steel mill cannot be converted to a wheat farm. Capital and land are *sector-specific*. This is the key departure from the Ricardian model.

Complete variable list:

L = total labour endowment

L_M, L_F = labour in manufactures, food
($L_M + L_F = L$)

K = capital stock (specific to M)

T = land endowment (specific to F)

Q_M, Q_F = output of M and F

P_M, P_F = prices of M and F

w = wage rate (same across sectors)

r_K = return to capital

r_T = return to land

Production Functions and Marginal Products

Production functions:

$$Q_M = F(K, L_M), \quad Q_F = G(T, L_F)$$

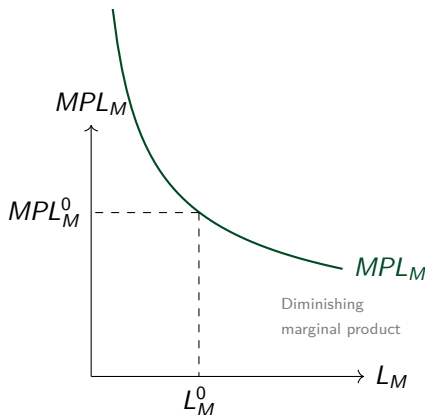
Both exhibit:

- **Positive marginal product of labour:**
 $MPL_M, MPL_F > 0$.
- **Diminishing marginal product:** as more labour enters a sector, MPL falls (holding K or T fixed).

Profit maximisation (per sector):

Under competition, each sector hires labour until
 $w = P_i \cdot MPL_i$:

$$w = P_M \cdot MPL_M(K, L_M)$$



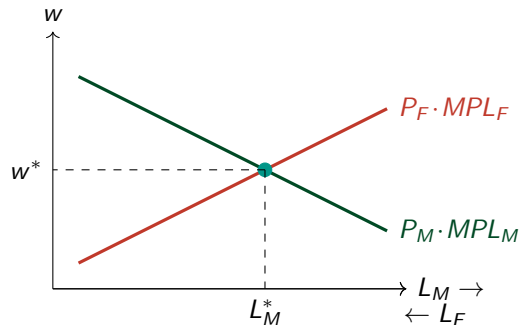
Labour Market Equilibrium: The Key Diagram

Labour market equilibrium: wages equate across sectors.

$$P_M \cdot MPL_M = P_F \cdot MPL_F = w$$

Reading the diagram:

- Horizontal axis: labour in M (left to right) and F (right to left). Total length = L .
- $P_M \cdot MPL_M$ slopes down (left to right): more labour in M lowers MPL_M .
- $P_F \cdot MPL_F$ slopes down (right to left): more labour in F lowers MPL_F .
- **Equilibrium:** crossing point $\Rightarrow L_M^*, w^*$.



Effect of Trade: P_M Rises (Home Exports Manufactures)

Suppose Home exports manufactures:

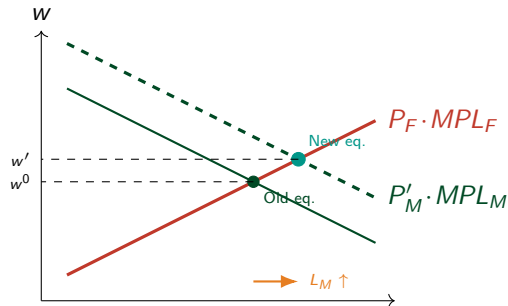
trade raises P_M .

Effect on labour market:

1. $P_M \cdot MPL_M$ schedule shifts **upward**.
2. At old wage, M sector now pays more $\Rightarrow$ workers move from F to M.
3. $L_M \uparrow$, $L_F \downarrow$.
4. Wage w rises, but **less than proportionally** to P_M (since MPL_M falls as more workers enter).

Key: wage rises by less than P_M rises.

Therefore: real wage in terms of M falls; real wage in terms of F rises.



Winners and Losers: Unambiguous Results

Effect of a rise in P_M :

Capital owners (specific to M) — WIN

$$r_K = P_M \cdot MPK_M(K, L_M^{\text{new}})$$

$P_M \uparrow$ directly raises revenue; more labour now in M raises MPK_M further. r_K rises *unambiguously* in **real terms**.

Landowners (specific to F) — LOSE

$$r_T = P_F \cdot MPT_F(T, L_F^{\text{new}})$$

Labour left food sector $\Rightarrow MPT_F \downarrow \Rightarrow r_T \downarrow$. Real return to land falls **unambiguously**.

Labour — AMBIGUOUS

Wage w rises, but by less than P_M rises.

- Real wage in M: $w/P_M =$ falls (wage rose less than P_M).
- Real wage in F: $w/P_F =$ rises (wage rose, P_F unchanged).

Net effect depends on consumption basket. Workers who spend mainly on manufactured goods are worse off; workers who mainly eat are better off.

Summary: Winners and Losers Table

Factor	Effect of $P_M \uparrow$	Reason	Political stance
Capital (M)	Gain	Specific to export sector	Favour free trade
Land (F)	Lose	Specific to import sector	Oppose free trade
Labour	Ambiguous	Mobile; depends on basket	Mixed
<i>Symmetric result: if $P_F \uparrow$ (Home exports food):</i>			
Land	Gain	Specific to export sector	Favour free trade
Capital	Lose	Specific to import sector	Oppose free trade

Political Economy Implication

Trade policy is contested because it creates *concentrated* gains for some and *concentrated* losses for others. The model explains why industries lobby specifically for protection of *their* sector.

The Magnification Effect (Jones, 1971)

Magnification Effect

Changes in goods prices are *amplified* into larger changes in factor returns. If P_M rises by $x\%$ and P_F is unchanged:

$$\hat{r}_K > \hat{P}_M > \hat{w} > \hat{P}_F > \hat{r}_T$$

where $\hat{X}$ denotes the percentage change in X .

Why magnification?

Capital owners get a **double gain**:

1. Direct price effect: P_M rises.
2. Indirect labour effect: more workers enter M (since wage rose), raising MPK_M further.

Landowners suffer a **double loss**:

1. No direct price gain (P_F unchanged).
2. Indirect labour loss: workers left F, lowering

Alberta Oil Boom 2002–2014

When oil prices tripled:

Factor	Effect
Oil capital owners	Enormous gains (+200%)
Alberta workers	Wage gains (+30–40%)
BC manufacturers	Lost workers; costs ↑
BC landowners (farms)	Hurt (Dutch disease)

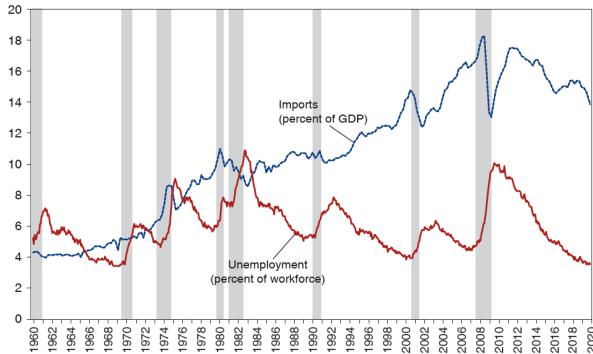
Does the Country Gain Overall?

- In aggregate, trade **raises national income** at world prices — the economy-wide pie grows.
- But the gains are *unevenly distributed*: specific factors in export sectors gain enormously; specific factors in import-competing sectors lose.
- **Could losers be compensated?** In principle yes (Kaldor-Hicks efficiency): the gainers' gains exceed the losers' losses.
- But compensation rarely occurs in practice — trade adjustment assistance programs typically cover only a fraction of losses.

The Political Economy Lesson

Concentrated losses (say, 50,000 steelworkers losing their jobs) are politically visible and generate intense lobbying. Diffuse gains (38 million consumers paying \$200 less for a car) are invisible. This asymmetry explains persistent protection.

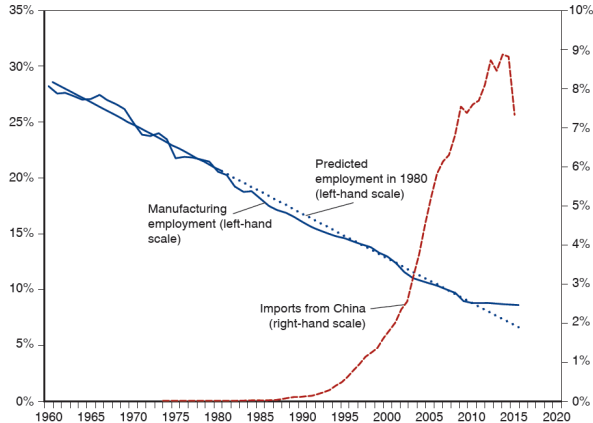
Figure 4-12: US Unemployment and Import Penetration



What this figure shows:

- **Horizontal axis:** import penetration ratio (imports/domestic consumption) in US manufacturing, by industry.
- **Vertical axis:** unemployment rate in that industry.
- Industries with higher import penetration have **higher unemployment**.
- This is the specific-factors model in action: workers in import-competing industries face job losses when trade opens.
- But note: *overall* unemployment does not rise with trade — workers eventually find other jobs (mobile factor adjustment).

Figure 4-13: US Manufacturing Employment and China Imports



What this figure shows:

- Plots **changes in US manufacturing employment** (horizontal) against **changes in Chinese imports** (vertical) across US regions.
- The negative relationship: regions exposed to more Chinese competition lost more manufacturing jobs (“China Shock”, Autor, Dorn & Hanson, 2013).
- The losses were **persistent**: workers did not quickly re-employ at similar wages in other sectors.
- This is evidence that labour is **NOT** perfectly mobile in the short run — the specific-factors model applies.

Policy Implication

Short Run vs. Long Run: A Bridge to Chapter 5

Feature	Short run (Ch. 4)	Long run (Ch. 5)
Capital	Specific to M	Mobile
Land	Specific to F	Mobile
Labour	Mobile	Mobile
Factors	3	2 (K & L)
Specialisation	Partial	Complete (tendency)
Winners/Losers	Clear	Via Stolper-Sam.

- Short run: capital is a machine that takes years to redeploy. Landowner cannot quickly convert farmland to a factory.
- Long run: given enough time, capital can depreciate and be re-invested elsewhere. Land can be rezoned.

Which Time Horizon Matters?

For policy analysis, both matter. Workers and capital owners care about the short run (their current jobs and assets). Economists often focus on long-run efficiency gains. The political conflict arises from this mismatch.

International Labour Mobility

Labour is relatively immobile internationally:

- Language barriers, family ties, legal restrictions, cultural adjustment.
- Nevertheless, ~280 million people live outside their country of birth.

Effects on host country (immigration):

- Labour supply $\uparrow \Rightarrow$ wages fall (workers competing for jobs).
- MPK_M and MPT_F rise (more labour to work with fixed capital/land) $\Rightarrow r_K$ and r_T rise.
- Net: capital and land owners gain; native workers may lose (especially low-skill).

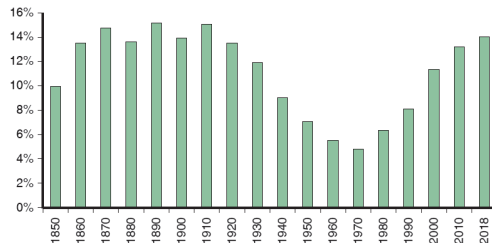
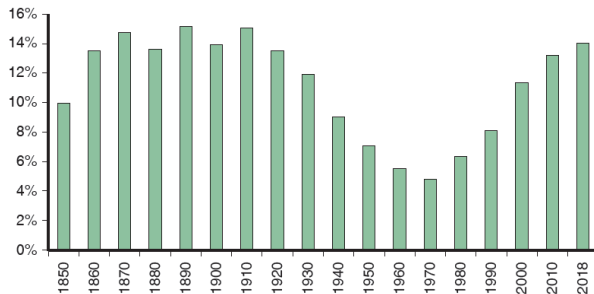


Figure 4-16: Foreign-Born Population as % of Total



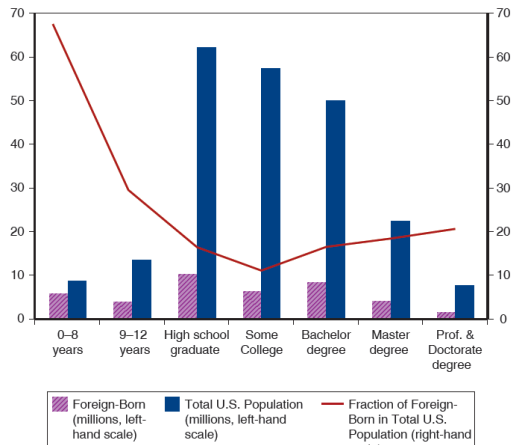
What this figure shows:

- Foreign-born share of the population across selected countries.
- **Switzerland, Australia, Canada:** very high (20–30%) — traditional immigration countries.
- **USA:** ~15% and rising historically.
- Higher immigration $\Rightarrow$ more labour supply $\Rightarrow$ specific-factors model predicts:
 - Capital/land owners gain.
 - Competing workers see wage pressure.

Canadian Example

Canada admits ~400K permanent residents per year (2023). Economic

Figure 4-17: Foreign-Born and Total US Population with Higher Education



What this figure shows:

- Compares educational attainment of foreign-born vs. total US population.
- Foreign-born workers in the US are **more educated on average** than native-born — they are not competing with low-skill workers, but with high-skill ones.
- **Two-tier immigration**: some immigrants are low-skilled (e.g., agricultural workers); others are high-skilled (H-1B visas for tech).
- Both types affect wages and returns to specific factors differently.

Implication

High-skill immigration raises returns

Key Takeaways

1. In the specific factors model, capital and land are sector-specific in the short run; only labour is mobile.
2. A rise in P_M **unambiguously benefits** capital owners (specific to M) and **unambiguously harms** landowners (specific to F).
3. Workers face **ambiguous** effects: wage rises but less than proportionally to P_M .
4. The **magnification effect**: factor returns change more than goods prices.
5. Countries gain in aggregate from trade, but distribution is unequal — creating political conflict.
6. The China shock evidence confirms that trade adjustment is slow and costly for workers in affected sectors.
7. International labour mobility creates symmetric distributional effects.

Next: Chapter 5 — Heckscher-Ohlin Model

Long run: all factors mobile. What determines trade patterns then?

Discussion

1. In the Canadian forestry sector, who gains and who loses when global lumber prices rise? Map to specific factors.
2. The US imposed tariffs on Canadian softwood lumber. Using the specific factors model, show who in the US gains and who in Canada loses.
3. If automation destroys manufacturing jobs, is this similar to import competition? How does the specific factors model apply?

Further reading:

- Jones, R.W. (1971), "A Three-Factor Model," *Trade, Balance of Payments and Growth*.
- Autor, D., Dorn, D., & Hanson, G. (2013), "The China Syndrome," *AER*.
- Borjas, G.J. (1995), "The Economic Benefits from Immigration," *JEP*.